programs → make decision
↓
conditional statements
↓
Control flow.

1. What is a condition?

→ if temp is above 40° → turn on the AC.
→ if exam marks > 35 → student pass else fail

T/F

condition always produces T/f.
print(10>5) → True.

{If}

if condition:
code block.
↓
indentation.

$\boxed{\text{SCE}}$

{if-else}

if age ≥ 18 → vote
else → cannot vote

if condition:
code
else:
code

Condition → True → if
→ else
more than two possibilities.

Ex:- Exam grading system

if + else

Marks ≥ 90 → Grade A
Marks ≥ 70 → Grade B
Marks ≥ 60 → Grade C
Marks ≥ 35 → Grade D
Else → Fail

} → elif

Virat
Anushka
Virushka

Syntax:

if condition:
code
elif condition:
code
elif condition:
code
else:
code

→ = nested if ==

1. if age = 18
if age == 18

2. indentation

3. (:) forgotten if condition:

Nested conditions.
↓
a decision inside another decision.

= if(——, T, if(——, T, f

college admission system:-

S1 → check if student passed exam
S2 → if passed → check marks
S3 → Based on marks → assign course.

IIT
CS: A2R → 63
CI
E
IIT Delhi

if condition1:
if condition2:
code

Print number from 1 to 20.

print(1)
print(2)
print(3)
⋮
print(20)

repetitive work.

loop → repeating a block of code multiple times.

While → repeat code as long as a condition is True.

While condition:
Code block.

before every iteration.

Break → to stop the loop immediately

if condition:
break

{Continue Statement}

↳ skips current iteration

{Pass statement}

pass → does nothing
It acts as placeholder.

Q1. Write a program using while loop to print numbers from 1 to 10.
Q2. Write a program to print sum of first 10 numbers.
Q3 " " " that stops printing numbers when it reaches 17.